AI Agents: Redefining Organizing

Abstract

As Artificial Intelligence (AI) evolves beyond a mere technological tool into autonomous organizational actors, we must reconceptualize our understanding of organizing. This point-of-view article challenges the prevailing human-centric narrative in organization design and argues for a need to challenge our existing insights in light of the arrival of autonomous AI agents. As AI agents develop capabilities akin to human agency, notably autonomously designing organizations, treating AI solely as a technology is no longer sustainable in organization design. This article outlines some of the theoretical implications and methodological challenges associated with studying future organizing. It also calls for interdisciplinary collaboration to advance existing insights and develop new ones to account for non-human-centric future of organizing.

Introduction

How is organizing changing with the advances in Artificial Intelligence (AI)? Even before the boom associated with Large Language Models (LLMs), which the public release of ChatGPT sparked in November 2022, AI had attracted the attention of organization design scholars curious to explore this frontier. Solely in the *Journal of Organization Design,* 14% of articles published between 2016 and 2021 referenced AI or “Artificial Intelligence.” This share has grown to 26% between 2022 and 2025 (early February). While scholarly attention toward AI has grown – and there appears to be a wealth of opportunities for future research – I argue that even this increased interest severely underestimates AI’s importance in organization design for both research and practice.

The reason for this claim lies in the prevailing human-centric narrative in organization design (see Mortimore 2024, for a discussion of its drawbacks). Currently, treating AI as mere technology (i.e., a tool that enables the fulfillment of human goals in a specifiable and reproducible manner, Brooks 1980) is a valid perspective. However, once efforts to design truly autonomous AI agents succeed, this perspective will quickly become outdated.

The term AI agent is used differently across the literature and practice; for our purposes, I adopt Wiesinger et al.’s (2024) definitions that view AI agent as “… an application that attempts to achieve a goal by observing the world and acting upon it using the tools that it has at its disposal… [it is] autonomous and can act independently of human intervention.”

Autonomy is the key feature of an AI agent and the primary reason for my proposal to renounce the human-centric narrative in organization design. Autonomy is about “leading one’s life according to reasons, values, or desires that are authentically one’s own” (Taylor 2014). Currently, the early versions of ‘AI agents’ remain technologies in the sense of tools for the fulfillment of human goals (Brooks 1980); they are not truly autonomous but rather able to act independently in limited tasks. For example, OpenAI recently unveiled Operator (OpenAI 2025), trained to interact with websites and perform a variety of tasks like booking accommodations or ordering groceries. However, technologies such as Operator are early steps in efforts to design agents that are autonomous in the sense described above – or even to grant them individual sovereignty (Hu et al. 2025), which is defined by ultimate freedom of actions, voluntary engagement in relationships, and non-interference of others with coercive methods (Ilievski 2015).

Rethinking Fundamentals of Organization Design in the Age of AI Agents

Scholarly attention in the organization design field has thus far focused on AI as a technology and how *human* organizing changes with novel technological tools. This approach follows a logic that has been used for other technologies that advanced organizing, and that has been conceptualized already in the early organization design literature (e.g., Chandler 1962, or Galbraith 1974). However, with AI agents transcending the traditional technology category, focusing solely on *human* organizing would provide an incomplete picture.

Subsequent renouncement of the human-centric narrative in organization design raises critical questions. In this article, I specifically focus on one of many: How will this shift reshape the fundamental problems of organizing (Puranam et al. 2014): the division of labor (dividing tasks and allocating them to actors) and the integration of effort (sharing information and distributing rewards)?

In the foreseeable future, we cannot treat AI in organizing as a single, undifferentiated entity. That is why we speak of AI agents. When multiple independent actors are involved, the division of labor remains a relevant problem. Moreover, the absence of a single entity implies the need for integration of effort. To further strengthen this point, consider two additional commonalities that AI agents will share with humans:

First, AI agents will exhibit bounded rationality (Jirásek 2024). Their capabilities will be necessarily limited by human influence in their original design, finite computational power, and imperfect information about the world. Thus, solving the problem of information provision remains essential for facilitating cooperation among agents.

Second, unlike traditional digital technologies, AI – especially built on advanced LLMs – incurs high marginal costs. Although these costs are expected to decrease dramatically over time (e.g., Altman 2025), the answers provided by top-class LLMs may currently cost thousands of dollars per task (Chollet 2024). This implies that AI agents will not always be available for any task; they will require rewards for the labor not only in the form of abstract optimization functions but also through tangible incentives, at least monetary compensation that covers the expenses of computing power (infrastructure and electricity).

In sum, the fundamental problems of organizing (Puranam et al. 2014) remain relevant in the future with autonomous AI agents.

In algorithmic organizing, illustrated by Decentralized Autonomous Organizations (DAOs), we have already seen that algorithms can address a considerable part of solutions to fundamental problems of organizing (see, e.g., Hsieh et al. 2018; Jirásek 2023). However, even in these cases, humans design and control changes to organizations (i.e., exception management(Puranam 2018), even if this is done collectively and in a decentralized fashion (like in the case of Bitcoin; Hsieh et al. 2018).

In contrast, with truly autonomous AI agents – agents that independently pursue their goals using the tools at their disposal (Wiesinger et al. 2024) – there is no fundamental reason why we cannot observe organizations that are designed and managed entirely by AI agents. In many cases, these organizations might primarily exist in the digital realm; yet, given the interconnections between physical and digital worlds (such as through integrated payment systems, communication networks, and Internet-of-Things sensors), there is no barrier to AI agent-designed organizations eventually manifesting in the physical world as well.

With the current state of the development, it is too early to observe such organizing outside experimental settings. One such early example is the Virtual Lab of Swanson et al. (2024), a team of diverse AI agents cooperatively designing new SARS-CoV-2 nanobodies. In the meantime, initiatives from large technology companies like OpenAI (with Operator as an early example; (OpenAI 2025) and smaller startups building on blockchain technology (e.g., Freysa.ai 2025) signal that efforts toward truly autonomous agents are underway.

Implications for the Field of Organization Design

Renouncing the human-centric narrative in organization design would lead to a need for reinterpretation or extensions of some theories, including foundational ones. For example, we have already discussed that AI agents are enabled by AI technology, and when deployed to fulfill human needs, they could be considered technology in Brook's (1980) sense, which aligns with a human-centric perspective. However, when these agents are endowed with autonomy – or even sovereignty (the former being the defining feature of agents, Wiesinger et al. 2024, while the latter aim of some initiatives, e.g., Freysa.ai 2025; Hu et al. 2025), it would be contradictory to see AI agents as technologies – or non-autonomous AI technologies (such as Robotic Process Automation, (Hofmann et al. 2020) as AI agents. Autonomy in this full-fledged sense is not just about operating without constant human oversight; it is about possessing the capacity to set, prioritize, and pursue goals independently, akin to human agency (like one discussed by Taylor 2014).

In some respects, AI agents will be qualitatively similar to humans: there will not be a single, undifferentiated entity, they will exhibit bounded rationality, and they will be reward-seeking. However, this does not imply that their limitations will mirror those of humans. For example, they will be able to replicate rapidly, process enormous amounts of data within moments, and perform many tasks at a fraction of human cost (while expensively struggle with others, (Chollet 2024).

While I have argued that the fundamental problems of organizing will remain relevant in the future, the solutions to these problems may diverge significantly from today's common categories. Future forms of organizing could be far more flexible, dynamic, and decentralized – capable of self-reconfiguration in ways that are impossible to foresee.

The implications of this shift for organization design research extend beyond theory. Traditional methodological approaches – surveys, interviews, ethnographic observations, and many simulations – are designed to capture human perceptions, decision-making, and social interactions. However, these methods may prove insufficient or even inadequate when organizing is driven by AI agents. AI agents operate based on algorithms and data, leaving digital traces rather than subjective narratives. This challenges our reliance on self-reported data (although LLM technologies might enable us to access AI "narratives") and human observations to understand the behavior of organizational actors and the emergent forms of organizing.

Interdisciplinary collaborations will thus become critical. For instance, partnering with computer and data science experts for algorithmic auditing and log analysis will be crucial for some studies. Approaches such as Computationally Intensive Theory Building (Miranda et al. 2022) will be relevant outside their home fields (information systems literature in this case). Such interdisciplinary collaborations will not only advance our theoretical understanding but also help address the wealth of ethics- and policy-related issues that the phenomenon of AI agents raises.

Conclusion

Autonomous AI agents represent a new frontier in organization design. Historically, technologies have advanced organizing (e.g., Chandler 1962; Galbraith 1974), and we saw an abundance of advancement enabled by digital technologies. For example, algorithmic organizing (e.g., Hsieh et al. 2018; Jirásek 2023) enabled considerable decentralization of organizations across space and size. However, these developments have not radically altered our conception of organizing. In contrast, I argue that autonomous AI agents will fundamentally transform organization design research and practice and necessitate a departure from the traditional human-centric narrative.

Although established organizational theories may still hold relevance, they will likely require reinterpretation or extension as AI agents increasingly function as actors on par with humans – and will be able eventually to design and reconfigure organizations, including those outside the digital world.

Moreover, moving away from a human-centric perspective introduces methodological challenges. Traditional methodological approaches – like surveys, interviews, and observations – may prove insufficient in AI-driven settings, underlying the need for rich interdisciplinary collaborations, particularly with computer science or information systems researchers.

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